

CMP-03 - Authority and Delegation Service

Own bounded authority, delegation, revocation, and policy-decision records without inferring authority from identity, trust, credentials, or capability.

Authority and state ownership

Book I: Decisions 1-7, 24-29, 35, 40, 47-50, and 58. Book II: Chapters 05. Authoritative state: Authority Grants, Delegation Records, revocations, Policy Decision Records, authority-context envelopes. Explicitly non-owned: authoritative Identity state, trust assessment, capability implementation, constitutional root authority.

Critical invariant

Every delegation is attributable, scoped, conditional, expiring, revocable, and bounded by the delegator's current authority.

Logical interfaces

CMP-03-IF-01 / Command / Grant Delegation

Command `Grant Delegation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-03-IF-02 / Command / Revoke Delegation

Command `Revoke Delegation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-03-IF-03 / Command / Evaluate Permission

Command `Evaluate Permission` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-03-IF-04 / Command / Expire Delegation

Command `Expire Delegation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-03-IF-05 / Query / Get Effective Authority

Query `Get Effective Authority` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-03-IF-06 / Query / Explain Policy Decision

Query `Explain Policy Decision` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-03-IF-07 / Query / List Active Delegations

Query `List Active Delegations` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-03-IF-08 / Event / Delegation Activated

Event `Delegation Activated` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-03-IF-09 / Event / Delegation Revoked

Event `Delegation Revoked` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-03-IF-10 / Event / Permission Decided

Event `Permission Decided` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Draft > Active > Suspended > Revoked or Expired. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: delegation cycle, scope escalation, stale revocation, policy-version mismatch, authority-context loss. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-03-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-03-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-03-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-03-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-03-TST-005 - Critical invariant. Every delegation is attributable, scoped, conditional, expiring, revocable, and bounded by the delegator's current authority. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-03-TST-006 - Failure containment. Inject delegation cycle, scope escalation, stale revocation, policy-version mismatch, authority-context loss. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-03-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-03-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-03-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-03-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.